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A multistate modeling approach for organizational cybersecurity exploration and exploitation

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ABSTRACT

This study examines the dynamic stages of exploration and exploitation efforts by organizations in their cybersecurity responses using multistate modeling. Using textual data from the annual 10-K reports of S&P 100 organizations, this study uses a combination of text analytics and Markov chain approach to quantify exploration and exploitation in organizational cybersecurity responses. The study models two and four states of exploration and exploitation based on their cybersecurity responses over time and uses a continuous-time Markov chain approach to analyze transitions between states as organizations adapt their responses over time to achieve ambidexterity. The two-state Markov model focuses on the firm-level Exploration and Exploitation states whereas the four-state model captures deeper levels of exploration and exploitation by considering Surviving, Investigating, Reinforcing, and Balancing as possible states for exploration and exploitation of cybersecurity responses. We analyze the dynamics of organizational exploration-exploitation behaviors by modeling longitudinal transition probabilities across different states. Implications for research and practice are discussed.

1. Introduction

Cybersecurity has become an increasingly important area for information technology (IT) and information systems (IS) research and practices over the last few years as several highly visible and impactful attacks on large and well-known organizations around the world have resulted in significant security breaches in recent years. Just to mention a few, Home Depot experienced data loss on 56 million cards, Equifax Corporation suffered a massive data breach for 145 million customers, Yahoo Inc., the giant Internet information company experienced a huge database breach with more than half a billion user accounts compromised, MasterCard Corporation announced one of the largest security breaches occurred on 40 million credit card users, and Marriott International declared a data loss on 500 million customers which have been publicized and become the center of attention [78–82]. Consequently, cybersecurity responses have become matters of significant concern to researchers and practitioners over time [1].

Prior research on cybersecurity is multi-faceted and addressed both organizational and operational concerns such as identification of threats or vulnerabilities to cybersecurity, mechanisms of cybersecurity breaches and attacks, social, economic, and financial impacts of data breaches, and routines, structures, capabilities for cybersecurity responses for organizations. Jenab and Moslehpour [2] reviewed the cybersecurity literature on various areas such as network security, information security, and cloud security. Gordon et al. [3] described the roles of leadership processes, organizational structures, and capital allocations through the installation of the Chief Information Security Officer positions within organizations for cybersecurity oversight and management. Cybersecurity breaches are known to influence market value, stock price, and market return of organizations negatively [4–8].

Organizational cybersecurity relies on the simultaneous pursuit of two activities: exploration and exploitation [9]. It can be conceptualized as an organization's ability to simultaneously explore new cybersecurity responses, resources, and practices and also exploit their current cyber security responses, resources, and practices [10]. However, a balance between exploration and exploitation is hard to achieve due to the contradictory nature of the two activities. If the organization has limited IT resources to support high levels of exploration and exploitation, exploitation is preferred over exploration. However, if the organization has ample IT resources to enable high levels of exploration and exploitation, exploration is prioritized over exploitation [11]. To date, little research has explored the dynamics by which organizations pursue cybersecurity exploration and exploitation. To address this research gap, we aim to answer the research question: *How do organizations explore or*

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exploit cybersecurity responses over time? We propose a modeling approach that can explain how organizations transition between exploration and exploitation states in handling cybersecurity responses.

Specifically, this research examines different types of transitions between exploration and exploitation in organizational cybersecurity responses using a novel analytical multistate modeling approach. Using textual data on cybersecurity responses extracted from annual 10-K reports released by S&P 100 organizations, text analytics and Markov chain analysis are used to empirically examine exploration and exploitation behaviors in the organizational learning process for cybersecurity. Particularly, this study uses a combination of text analytics and Markov chain modeling approach to understand how organizational cybersecurity responses evolve through periods of stability (exploitation) and change (exploration) with different phases of surviving, investigating, reinforcing, and balancing. Mathematically, such organizational shifts to achieve ambidexterity over time can be interpreted as a continuous-time Markov chain model. Our proposed Markov chain model uses text-mined data and formulates the behavior of an organization as a sequence of possible states (e.g., surviving, investigating, reinforcing, and balancing) where the organization begins in one of the states and moves consecutively to another state in the face of changed market conditions. While the stochastic nature of the Markov model allows us to accommodate variety-seeking behaviors in organizational cybersecurity responses, it can also capture the central tendency of the observations which can be used to test for the stability of the pattern of responses over time.

This study, therefore, has three main contributions. First, we contribute to the organizational literature by proposing a novel analytical modeling approach for studying exploration and exploitation in the organizational learning process. Specifically, we develop continuous-time Markov models to demonstrate that organizations alternate between periods of exploration and exploitation to achieve ambidexterity over time. Second, we contribute to the organizational cybersecurity literature by identifying stages through which organizations transition to achieve a balance between exploration and exploitation in their cybersecurity response portfolios. Third, we contribute to the business analytics literature by demonstrating how textual features extracted from organizational reports via text mining techniques can be used in a different analytical approach (i.e., Markov chain). We illustrate how a combination of advanced analytics techniques can result in a better understanding of such a complex organizational phenomenon as exploration-exploitation.

The remainder of the paper is organized as follows. The next section introduces the related literature and theoretical background informing this study. The research model and research methods are described in the subsequent two sections. These are followed by the results and discussion sections, and the conclusion section wraps up the paper.

2. Theoretical background

2.1. Cybersecurity response

Cybersecurity response represents the actions undertaken by organizations to prevent, monitor, detect, mitigate, and manage cybersecurity threats [12,13]. Such responses are aimed at protecting organizational computing resources including data, hardware, software, and networks from unauthorized access, disruption, damage, theft, loss, or malicious attack [14,15]. Organizations may implement IT and non-IT countermeasures to secure their physical facilities, personnel, and computing resources [16]. IT countermeasures may include encryption, firewalls, antivirus software, anti-malware, secured servers, and intrusion detection software [17–19]. Non-IT countermeasures may comprise training users on proper protocols, properly disposing of media containing sensitive data, building redundancies to recover data, and securing physical locations [20–22].

The nature of threats has changed over time to include viruses, malware, sniffing, eavesdropping, phishing, structured query language injections, cross-site scripting, denial-of-service, and Trojans [23,24]. Consistent with such changes, the cybersecurity profession has developed different protocols and best practices to monitor, identify, and counter threats. Cybersecurity professionalization has thus resulted in new education and training programs that may be required of cybersecurity professionals employed within organizations. Moreover, organizations need to comply with cybersecurity requirements specified by governmental directives and regulations (e.g., Health Insurance Portability and Accountability Act: HIPAA, Sarbanes-Oxley Act: SOX, and Federal Information Security Modernization Act: FISMA) or industry standards and practices (e.g., Computing Technology Industry Association: CompTIA, Payment Card Industry Data Security Standard: PCI-DSS) [25–28].

Due to the diversity and sophistication of techniques employed in cybersecurity attacks, the changes in industry norms and practices, and the need for regulatory compliance in modern times, organizations need to plan, design, and implement safeguards and countermeasures continually. This implies that organizations cannot always rely on existing cybersecurity solutions (which may have been appropriate at an earlier time) but proactively develop new and relevant solutions (consistent with the changing threats in the current environment). However, organizations may not be able to successfully identify such newer solutions immediately but spend time experimenting with various options before finalizing specific measures that could be subsequently utilized. Such transitions between experimenting with and utilizing new solutions are found in the exploration/exploitation thesis in organization literature [29].

2.2. Dynamics of organizational exploration–exploitation behaviors

It is well accepted that organizations need to engage in exploration and exploitation to survive, sustain and grow in a changing environment [29]. Exploration refers to behaviors that include risk-taking, experimentation, and innovation to discover new knowledge or replace existing knowledge within an organization [30,31]. In the cybersecurity area, this involves exploring and switching between different strategies to create new procedures or solutions and expand existing capabilities to include new or alternate responses to cybersecurity threats. Exploitation represents organizational behaviors that refine, implement, and execute existing knowledge in an organization to secure efficiency. It leverages existing knowledge, procedures, technologies, and solutions to maximize organization benefit [32–34]. Prior research concluded that exploration and exploitation can generally improve firm performance [35,36].

Organizations often engage in varying combinations of exploration and exploitation over time. Jeyaraj and Zadeh [10] identified four possible states to distinguish between low and high levels of exploration and exploitation. When both exploration and exploitation are low, organizations operate in a Surviving mode. A high level of exploration with a low level of exploitation is considered as Investigating, whereas a low level of exploration but a high level of exploitation is considered as Reinforcing. When both exploration and exploitation are high, organizations are considered as balancing or ambidextrous because they perform these rather opposing activities simultaneously. Fig. 1 displays the four possible states for exploration and exploitation which will be used as a basis to perform a four-state Markov chain analysis.

Prior literature addressed structures [37], contexts [38], and leadership processes that enable organizations to perform exploration and exploitation simultaneously [39]. Azoulay-Schwartz et al. [40] studied exploration vs. exploitation in the context of electronic commerce. Duncan [41] suggested that organizations sequentially accomplish ambidexterity by shifting organizational focus over time. That is, for organizations to accommodate the conflicting alignments required for exploration and exploitation, they need to shift their structure over time to adapt to environmental and technological changes [37].

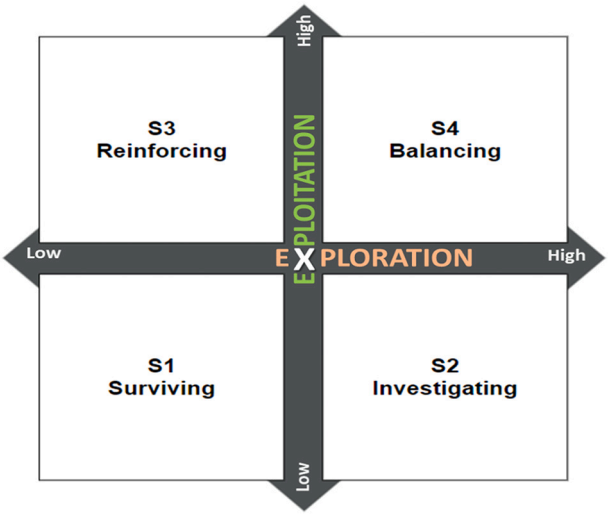

Fig. 1. Exploration vs. Exploitation States.

Organizations take two distinct approaches to implement a temporal transition or shift between different levels of exploration and exploitation over time [42]. The first approach conceptualizes temporal transition as discontinuous jumps between exclusive periods of exploration and exploitation [43,44]. This is grounded in the punctuated equilibrium theory which argues that organizations engage in stable periods of exploitation interspersed with sporadic episodes of exploration. The second approach views temporal transition as gradual or progressive changes between exploration and exploitation over time [45–47]. This is supported by multiple organization theories which suggest that organizations are resistant to change but can change only incrementally by engaging in continuous process of improvement [35]. Table 1 provides a summary of the extant literature on temporal transition between exploration and exploitation.

In this study, we demonstrate that organizational cybersecurity responses evolve through periods of stability (exploitation) and change (exploration) with different phases of surviving, investigating, reinforcing, and balancing. Mathematically, we propose that such organizational shifts to achieve a dynamic balance between exploration and exploitation over time can be interpreted as a continuous-time Markov chain model. The Markov chain is a class of stochastic processes that provides a simple mathematical model yet sophisticated enough, equipped with a well-known family of statistical tests [49–51], to understand the behavior of a system (e.g. organization) over time [52]. It formulates the behavior of a system as a sequence of possible states (e.g., surviving, investigating, reinforcing, and balancing) where an organization begins in one of the states and transitions consecutively from one state to another to reflect changed environmental conditions or strategies.

In the product development domain, exploration and exploitation variables are obtained from their respective performance. The exploration variable is created based on the percentage of turnover that is due to products that are new to the organization or new to the market. The exploitation variable is constructed with the same approach, using the percentage of turnover that is due to products that do not undergo any alteration [53]. We use the same approach to quantify exploration and exploitation in the cybersecurity domain. Our exploration metric reflects the percentage of cybersecurity responses that are new to the firm, whereas the exploitation metric indicates the percentage of cybersecurity responses that did not change over time.

2.3. Markov chain models

A Markov chain is a stochastic model that describes the behavior of a system (e.g. individual, organization) over a period of time [54]. It identifies the underlying processes through which a system transitions from one state to another. Such models have been successfully applied to various business problems including predicting customer behaviors [55,56], customer lifetime value [57], online health infomediary [58], supplier selection [59], cloud services selection [60], and search engine optimization [61]. However, the use of a Markov model to characterize exploration and exploitation behaviors in the organizational learning process is new and is the focus of this research paper. This research uses a multistate Markov model to identify stages that organizations transition through in their efforts to balance between exploitation and exploration.

3. Research methods

Our research methodology includes data collection, preparation, modeling, and analysis which are described in the subsections below. The overall research methodology is shown in Fig. 2.

3.1. Sample

The top 100 organizations listed in the Standard & Poor's (S&P) index between 2014 and 2020 were reviewed for inclusion in the analysis. After a closer examination of S&P 100 lists, it was determined that the membership of organizations in the rankings is subject to change over time. Therefore, the combined S&P 100 rankings table was queried to find only the organizations that had stayed on the S&P list for seven years in a row from 2014 to 2020. This process concluded 87 organizations in common between the 2014–2020 rankings. Appendix A includes the details of the organizations in our sample.

3.2. Data collection

According to the Securities and Exchange Commission (SEC), public organizations are required to release cybersecurity matters in their annual reports, generally under the risk discussion section [62]. Prior studies have extracted substantial information about organizational processes and practices from annual reports [63]. Wang et al. [64] conducted a content analysis on the risk factors section in public filings such as 10-K reports and concluded that such documents contain action-oriented terms such as "implement" and "prevent" to disclose a firm's efforts to adopt precautionary measures or responses associated with cyber security.

In this study, the 10-K annual reports were used to capture textual data related to cybersecurity as disclosed by the organizations. An R-scraper, using the *EdgarwebR* package, was developed to retrieve the 10-K annual reports from the SEC Edgar database.[1] Specifically, 609 10-K reports were downloaded for the 87 S&P organizations in our study for the years 2014 to 2020.

3.3. Data preparation

SAS Text Miner was used to analyze unstructured data extracted from the 10-K reports and convert them into numerical data for statistical analysis with Markov chain models. As shown in Fig. 2, several natural language processing (NLP) techniques such as tokenization, dictionary-based stemming, part-of-speech recognition, n-gram, and synonym replacement were applied to accomplish this task. Parsing and tokenization algorithms were applied to the textual data from the 10-K reports to create a document-term matrix (DTM) that can be used for Markov chain analysis. Then, common text filtering techniques were used to clean the DTM from non-alphanumeric symbols such as non-standard characters, punctuation, and stop words. Furthermore,

[1] https://www.sec.gov/edgar/searchedgar/companysearch.html

Table 1
Prior research on exploration and exploitation.

Study	Discontinuous Transition	Incremental Transition	Focus	Data	Method
Burgelman [48]	✓		CEO	Interview, annual reports, other archives	Case study
Rothaermel and Deeds [45]		✓	R&D	BioScan	Regression
Lavie and Rosenkopf [46]		✓	Alliance Formation	Compustat	Cross-sectional time series regression
Mudambi and Swift [43], Swift [44]	✓		R&D	Compustat	Regression
Raisch and Tushman [47]		✓	New Business Initiatives	Interview, archives	Case study
Kang and Kim [35]		✓	Technological Innovations, patents	Compustat, CRSP, Patent database	Regression
Liang, Wang [11]		✓	IT	Survey	Structural equation modeling
Current Study		✓	Cybersecurity	Annual reports, Breach archive	Text mining, Markov chain

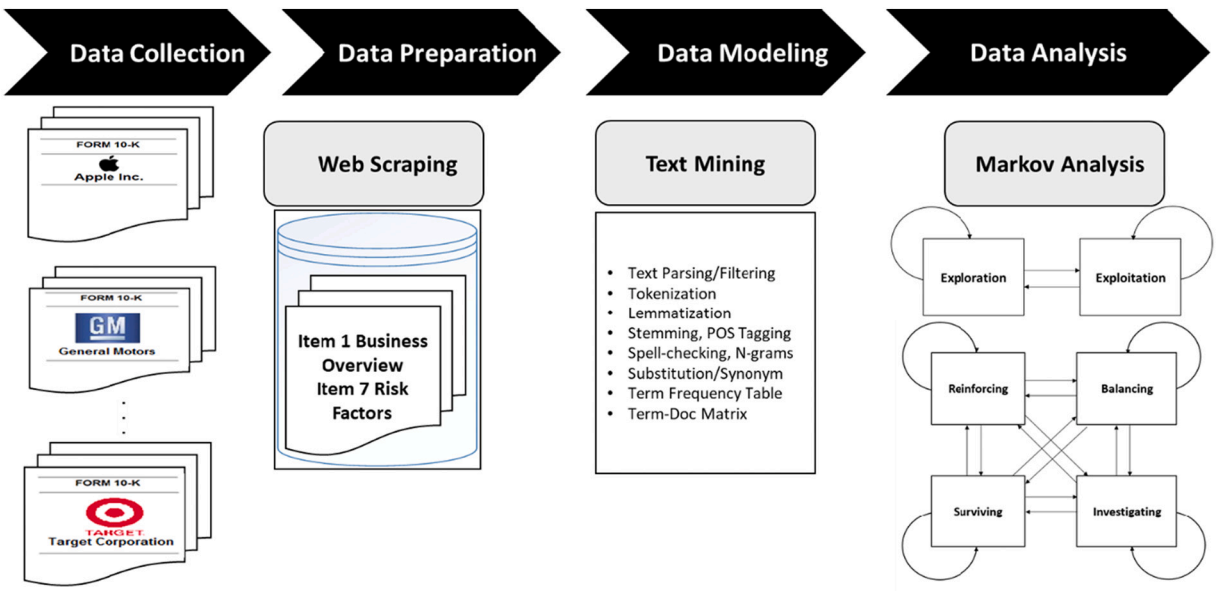

Fig. 2. Research Framework.

morphological analysis via dictionary-based stemming (aka lemmatization) was performed to consolidate different variations of a term to one single root term (e.g., stem) and generate only valid terms as stems. Next, part-of-speech (POS) tagging was done in such a way that the same terms with different POSs were treated as different terms in the term selection process. This allowed us to recognize the different ways terms were used and exclude those including auxiliary verb, conjunction, particle, preposition, and pronoun as parts of speech. To handle multi-term words or phrases, the 'noun-group' option in SAS Text Parsing was utilized to perform n-gram (i.e., 1 = unigram, 2 = bigram, 3 = trigram) analysis, look for patterns, and identify groups of words that should be processed as single terms. Finally, abbreviations, words, and phrases on the term table that carried the same meaning were identified and combined for a consistent representation of cybersecurity responses across all organizations in our sample. For example, the chief information security officer and CISO were treated as synonyms in the term table. This process yielded a cleaned DTM which contained the frequency of occurrence of each term as they appeared in the 10-K reports.

3.4. Data modeling

A set of terms and phrases that describes an organization's response to the cybersecurity threats was compiled and used to capture the ways in which organizations may explore or exploit cybersecurity measures in an environment (see Table 2 for the complete list of keywords [65]). The keywords were reviewed and refined over several iterations. These terms were used in the text mining process to find the frequency of the cybersecurity response by each organization. Since some of the terms and phrases for cybersecurity response had a low frequency of appearance, such keywords were not filtered out from the DTM by adjusting the

Table 2
Terms and phrases used for cybersecurity response quantification.

Keywords
access control, antivirus, audit trail, authentication, authorization, chief information security officer, chief security officer, CISO, cloud, cryptography, CSO, cybersecurity, destroy, disable, eavesdropping, exploit, expose, firewall, forensics, intrusion detection, scanner, screen scraper, secure gateway, security, security training, surveillance, two-factor identification, multifactor authentication, violation, encryption, compliance, payment card industry data security standard, PCI DSS, system and organization controls report, SOC, cloud controls matrix, CSA CCM, identity and access management, single sign on, disk striping, data encryption, disk mirroring, masking, warm site, cold site, hot site, regulations, policies, programs, plans, trainings, business continuity, disaster recovery, security controls, evaluation, testing, monitoring, updating, reporting, assessment, auditors, audits, alerting, logging, detecting, backup, entitlement, federated identity management, auditability, traceability, accountability, network security, privacy liability, cyber liability, regulatory, security, cyber security, cyber, sensitive data, sensitive information, sensitive personal information, confidential information, confidential data, personal data, proprietary data, proprietary information, intellectual property protection

cutoff threshold parameters in the text parsing and filtering algorithms. This process produced a cleaned term-document frequency matrix (TDFM) that was input in the Markov chain analysis.

Two separate metrics were created and applied to the term-document frequency matrices to measure exploration and exploitation. Suppose X_{ij} is the number of cybersecurity responses for organization i for year j, and C_i is the ratio of number of years in which number of cybersecurity responses (i.e., X_{ij}) changed from prior year (i.e., X_{ij-1}). The exploration metric for organization i as follows:

$$Explore_i = \overline{X}_i{}^*C_i \; i = 1\ldots87$$

where $\overline{X}_i$ is the average number of cybersecurity responses across all years (2014–2020).

Suppose Y_{ij} is number of years cybersecurity response i continued for several years for organization j, and D_i is the ratio of number of cybersecurity responses (i.e., Y_{ij}) that continued for several years. The exploitation metric for organization i as follows:

$$Exploit_i = \overline{Y}_i{}^*D_i \; i = 1\ldots87$$

where $\overline{Y}_i$ is the average number of years for every cybersecurity response for organization i. These two formulas were used to score organizational cybersecurity exploration and exploitation. While the former reflects the extent to which the firm adopts new cybersecurity responses over time, the latter indicates the extent to which the firm dealt with the same cybersecurity responses over time. The Z-score method was used to normalize the exploration & exploitation scores. The normalized exploration and exploitation scores were then used to determine the states occupied by each organization in each of the time periods under study. The states for all 87 organizations were then input into the Markov model (described next) to collectively understand the dynamics of exploration-exploitation behaviors in organizational learning for cybersecurity. Appendix A includes the state sequence for all the organizations in our sample during five time periods (ending 2016, 2017, 2018, 2019 and 2020).

3.5. Multistate Markov chain model

A multistate Markov chain is often used to explain a stochastic process in which an entity (e.g. organization) transitions through a discrete number of finite states [66]. In this model, future states that the entity attempts to reach depend on the current state and not on the previous ones. Depending on how time is treated, a Markov chain model can be discrete or continuous. A discrete-time Markov chain model assumes that state changes take place through a series of discrete-time steps which are known and observable. However, in a continuous-time Markov model, it is assumed that the exact sequence of state transitions may not be observable at the time of recording but can be inferred from the data [58].

In the context of this study, a continuous-time Markov chain is more suitable because we can only observe an organization's exploration-exploitation level through their 10-K reports, but the exact timings on when state transitions have occurred are hidden. The multistate continuous-time Markov chain model built in this study is used to understand the dynamics of organizational exploration-exploitation behaviors. A continuous-time Markov chain model is formulated by a set of transition intensities where each entry represents the instantaneous intensity of transitioning from one state to another. This study uses two types of Markov models which are shown in Fig. 3. Model I captures the high-level exploration and exploitation stages whereas Model II captures a deeper level of states: surviving, investigating, reinforcing, and balancing.

We assume that an organization transitions through a series of states over time. We rely only on yearly observed data to determine an organization's transition from state to state. We analyze seven years of data with year as the unit of analysis as it is reasonably short but long enough to observe the dynamics of an organization's exploration and exploitation behaviors. We observed each organization's state for each year from the data captured for exploration and exploitation through the annual 10-K reports.

Let $S_i(t)$ denote the state occupied by organization i at time t. Model I assumes that an organization resides in one of the following states based on its exploration and/or exploitation activities for cybersecurity responses: $S_i(t)$= State in year t for organization i, where $S_i \in \{Explr, Explt\}$.

In Model II, because organizations can engage in different levels of exploration and exploitation over time, we further categorized exploration and exploitation into four distinct states (surviving, investigating, reinforcing and balancing): $S_i(t)$= State in year t for organization i, where $S_i \in \{Survive, Investigate, Reinforce, Balance\}$.

To model the transition states, we captured organizations' exploration and exploitation activities for each year through their annual reports. We did not include "absorbing" states as organizations may move to any state from any other state. The arrows in Fig. 3 show the possible transitions between states. For example, organization i's state at time $t+1$ ($S_i(t+1)$) either stays the same as time t ($S_i(t)$) or moves to another. Both models allow the transitions from any state to the other as no absorbing state exists.

In a continuous-time Markov process with discrete states, initial transition probabilities $p_{rs}(t)$ are defined from the observed data. $p_{rs}(t)$ gives the probability of an organization being in state s at time $t+\Delta t$ in the future given that an organization is in state r at time t [66]:

$$p_{rs}(t) = P\{S_i(t+\Delta t) = s \,|S_i(t) = r\}$$

where $0 \le p_{rs} \le 1$, and $\sum_{\forall s} p_{rs} = 1$. The transition probabilities (P) are obtained from the observed data by estimating the transition rates over short time intervals depending on the availability of the data (e.g., yearly). Transition intensities (Q) are then calculated from the transition probabilities.

$$q_{rs} = \lim_{\delta t \to 0}\left(\frac{p(S_i(t+\delta t) = s|(S_i(t) = r}{\delta t}\right)$$

where q_{rs} represents the instantaneous transition rate of moving from state r to state s. Also, to ensure that at least one state must be occupied at time $t+\delta t$, the entries on the main diagonal of matrix Q i.e. q_{rr} are defined as $q_{rr} = -\sum_{s\neq r} q_{rs}$, that is, the sum of each row entries is zero. Due to the Markov property, the occupancy time in a state, known as sojourn time, is assumed to be exponentially distributed. This allows the underlying stochastic process to have a statistically equivalent discrete state representation.

A maximum-likelihood estimation through numerical optimization is employed to estimate all the parameters of the models included in the transition intensity matrix Q [67]. Given the observed states for organization i during the time $t_1, t_2, \ldots, t_n$ i. e. $S_i(t_1)S_i(t_2)\ldots S_i(t_n)$, the likelihood function for organization $i$is formulated as follows:

$$L_i(Q) = \prod_{k=1}^{n} p_{s_i(t_k)s_i(t_{k+1})}(t_{k+1}\text{-}t_k;\, Q).$$

The full likelihood is then formulated as the matrix-product of probabilities of transition between observed states, over all organizations i and observation times k:

$$L(Q) = \prod_i L_i = \prod_{\forall i}\prod_{\forall k} p_{s_i(t_k)s_i(t_{k+1})}(t_{k+1} - t_k; Q)$$

Once the transition intensity matrix Q is estimated, the transition probability matrix P can be obtained by taking the matrix exponential of the transition intensity matrix Q:

$$P(t) = e^{Q(t)}$$

We also calculate the mean average time for each state (or mean sojourn time). The mean sojourn time for state r is $1/q_{rr}$ where q_{rr}

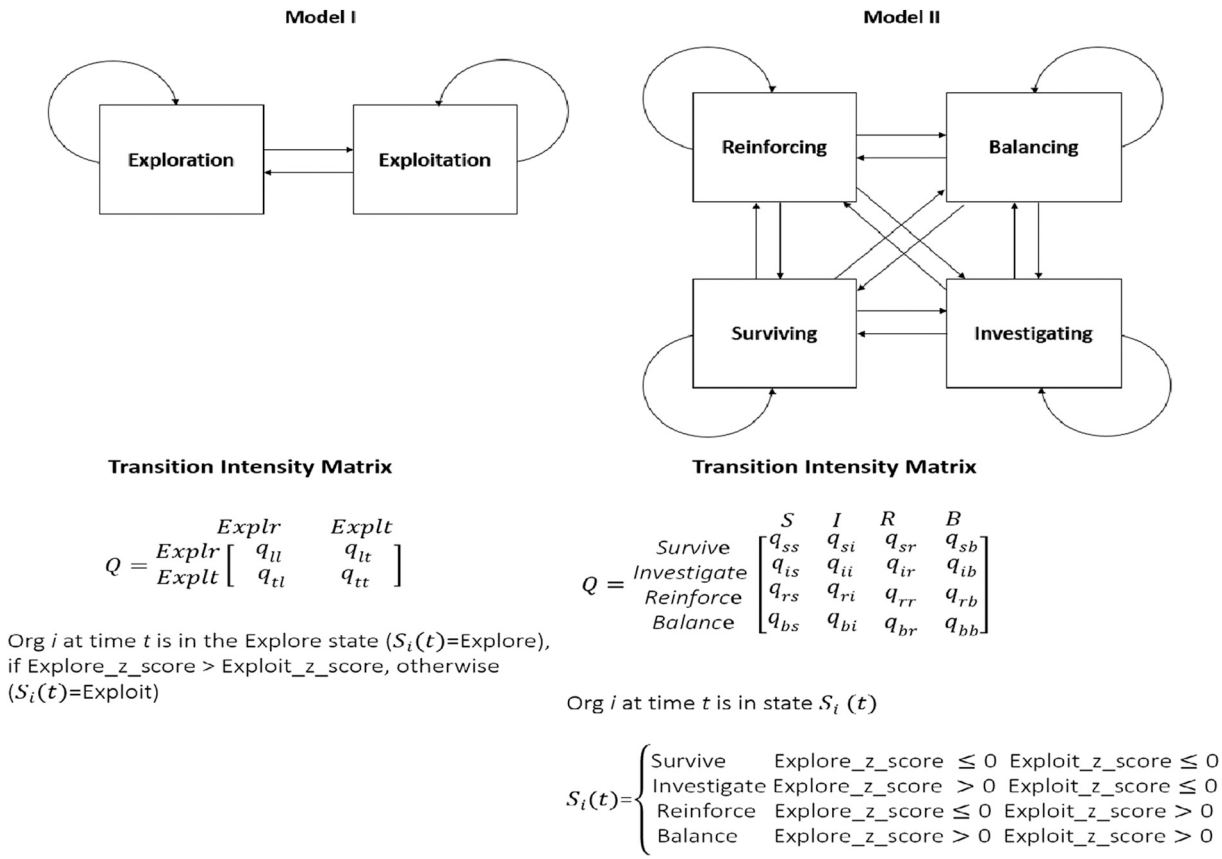

Fig. 3. Multistate Markov model for organizational exploration and exploitation efforts.

represents the *r-th* diagonal entry of the estimated transition intensity matrix Q.

These matrices allow us to examine organizational cybersecurity responses over time–collectively and individually–by simulating the state distributions and predicting the next state of the cybersecurity response considering the current state. The R-package *msm*, Multistate Markov and Hidden Markov Models in Continuous Time, was used to fit Markov models to the observed data using the Maximum-likelihood estimation described above and determine the parameter estimates [68].

4. Results

4.1. Main analysis

Based on the criteria described in Fig. 3, we identified 236 exploration states (54.2%), and 199 exploitation states (45.8%) over the time period 2016–2020 for Model I. For Model II, the exploration and exploitation states are further categorized into 102 surviving states (23.4%), 111 investigating states (25.5%), 125 reinforcing states (28.7%) and 97 balancing states (22.2%). A total of 421 transitions were recorded in Model II (60.4%).

According to Model I, there are different types of transitions which involve varying levels of intensity in exploration and exploitation. When organizations are in the exploration (exploitation) state, they can continue to explore (exploit) or they can stop exploring (exploiting) and start exploiting (exploring). Fig. 4 presents the transition probabilities for the two-state model in two transition periods—from the first year (2016) to the second year (2017) and from the first year (2016) to the final year (2020) for ease of explaining the results.

The results indicate that in a short run, the majority of organizations remained in the same state that they were in before. 85% of organizations in the exploration state during the first year continued to explore in the subsequent year whereas 15% of the organizations stopped exploring and started exploiting in the subsequent year. Similarly, organizations in the exploitation state showed a high transition probability of continuing to stay in exploitation state. As shown in Figs. 4, 91% of these organizations remained in the exploitation state with only 9% moved to the exploration state.

Although most organizations remained in status quo over a short period of time, they changed behaviors in the long run. As Fig. 4 shows, the transition probabilities to the same state dropped significantly by the end of 2020 with only about 53% of organizations in the exploration state continuing to explore while 46% transitioned to the exploitation state. Similarly, 71% of organizations engaged in exploitation initially continued with it to the end but 28% of the organizations changed to exploration to balance their efforts. The estimated mean sojourn times in exploration *and* exploitation states are 5.71 (95% CI[2] 3.84–8.50) and 9.43 years (95% CI 5.83–15.25) respectively. Thus, organizations are more likely to stay in an exploitation state than an exploration state. We also examined the effect of industry sector by allowing covariates in the Markov models. Appendix B shows the transition probabilities by industry sector.

The four-state Markov model was established to examine the underlying mechanisms in state changes between *exploration and exploitation*. Fig. 5 and Fig. 6 present the short-term and long-term transition probabilities between the sub-states in the four-state model.

First, in the short period of time (see the transition probabilities from 2016 to 2017 in Fig. 5), most organizations tend to remain in the same state in the next time period. This is evident from p_{ii} values which are significantly larger than the other transition probabilities i.e. $p_{ij}(i \neq j)$.

[2] Confidence interval

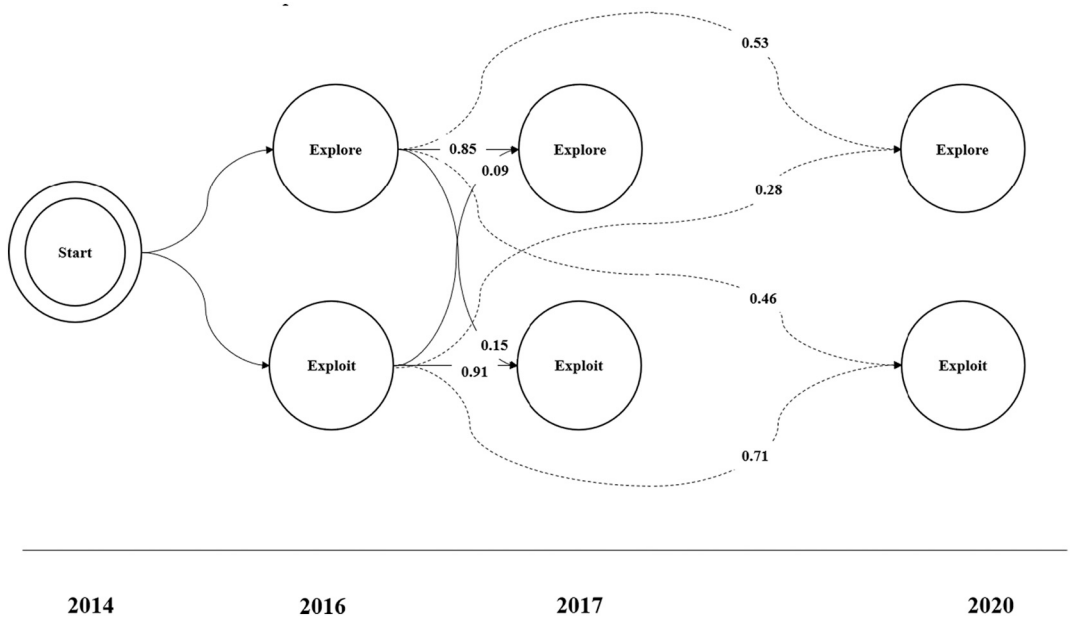

Fig. 4. Two-state Markov model transition probabilities (Model I).

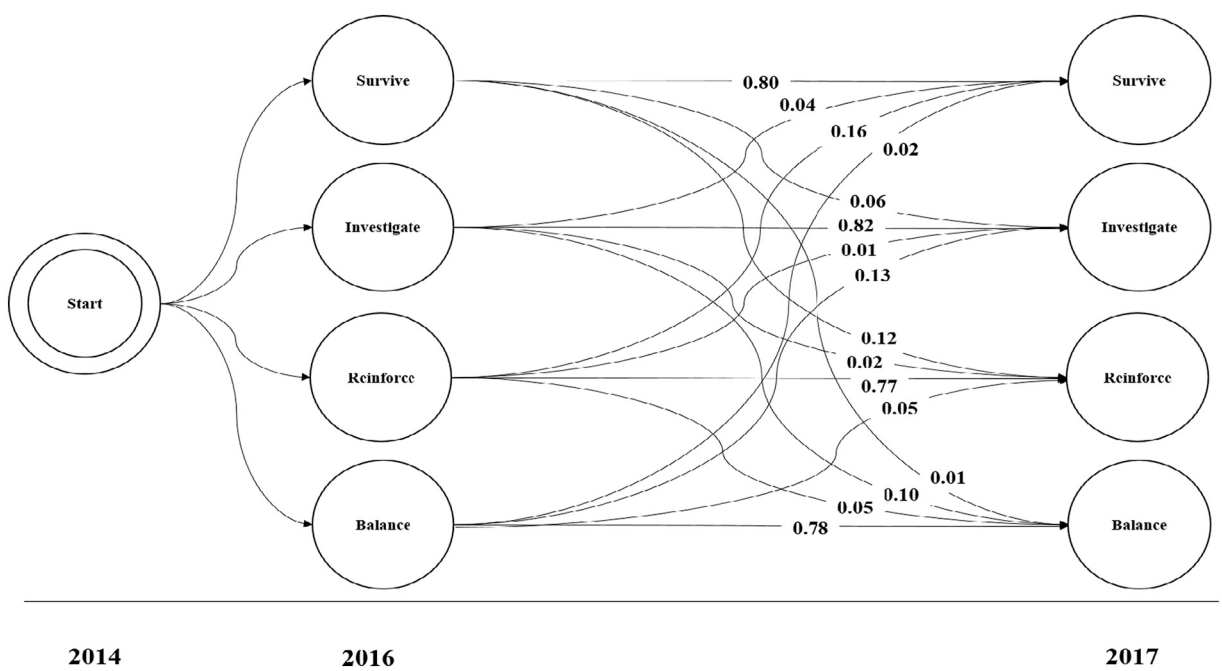

Fig. 5. Four-state Markov model with one-year transition probabilities (Model II).

Organizations in the investigating state seem to have the highest probability of staying (82%) than those with surviving (80%), reinforcing (77%) or balancing (78%) activities. In addition, the probability that an organization in the surviving state leaves its state in the following year is only 20%, whereas it is 18%, 23% and 22% for the organizations in the investigating, reinforcing and balancing states, respectively. Further, the very low transition probability between surviving and balancing states and vice versa suggest that organizations do not engage in discontinuous shifts between drastically opposed and conflicting procedures and capabilities. This implies that organizations prefer upgrading or downgrading their exploration and exploitation efforts by transitioning through intermediary states of reinforcing and investigating. Moreover, the transition probabilities from the balancing state to the investigating and reinforcing states are 13% and 5% which indicate that organizations could go back to these states in the next year to pursue a one-sided approach in their exploration and exploitation strategies. The transition probability from investigating to reinforcing (2%) is slightly greater than the other way (1%) which could intuitively imply that investigating is precursor to reinforcing. Finally, considering possible shifts from investigating, reinforcing, and balancing states to the surviving state, it can be argued that lowering exploration does not necessarily increase exploitation or vice versa as organizations may find themselves in a surviving mode and conduct minimal exploration and exploitation.

Over the long run of five years (refer to the year 2016 to year 2020 transition probabilities in Fig. 6), the transition probabilities between the same states (p_{ii}) dropped down considerably indicating that

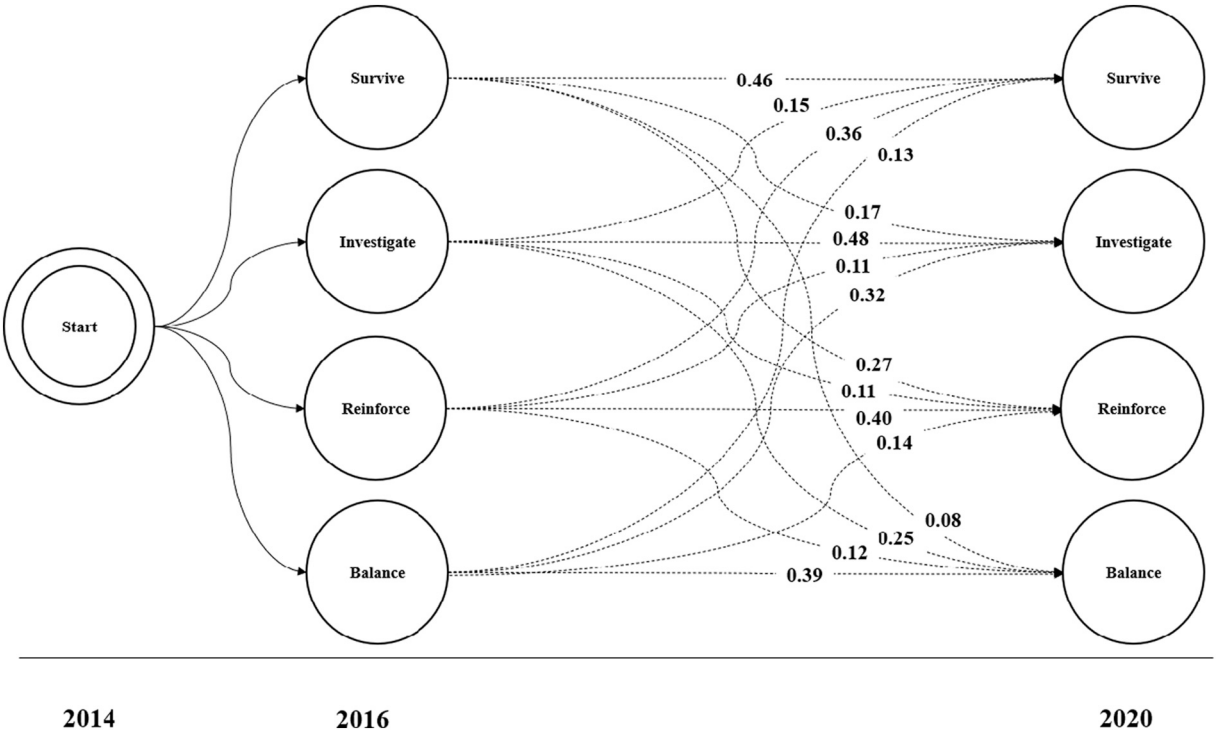

Fig. 6. Four-state Markov model with five-year transition probabilities (Model II).

organizational strategies on cybersecurity change over time. Considering transition probabilities from the surviving state to other states, organizations have a higher chance of reinforcing (27%) than investigating (17%) or balancing (8%).

The highest probability from the investigating state to another state over the long term is the balancing state (25%). It is also possible that these organizations find themselves in the surviving state trying to limit their exploration efforts (15%) or in the reinforcing state trying to engage in a greater exploitation (11%). While being at the reinforcing state initially, it is possible for organizations to remain in the same state of reinforcing (39%) or reach the balancing and surviving states with 12% and 36% probabilities respectively.

Different patterns can be observed for the balancing state. First, the balancing state has the lowest probability of remaining in the same state among others which suggests that organizations in the balancing state more often go back to the other states to explore or exploit solutions that they may not have identified during their times in those states. Second, while it is possible for organizations to switch gears back to the investigating state to engage in knowledge exploration mode and search for new cybersecurity responses or to the reinforcing state to apply the existing knowledge, they could also completely downgrade and find themselves in a surviving mode in the face of changed market conditions. Third, based on the Markov property of the models, the balancing state is not an absorbing state which means the balancing is not a "once and done" stage and organizations may have to leave the balancing state to realign their focus on exploration and exploitation in response to the growth and sophistication of cybersecurity threats.

The estimated mean sojourn times in surviving, investigating, reinforcing *and* balancing states are 4.20 (95% CI 2.55–6.94), 4.83 (95% CI 2.93–7.94), 3.60 (95% CI 2.39–5.42), and 3.90 (95% CI 2.41–6.33) years respectively. These suggest that organizations on average spend approximately 4 years in a state before leaving that state.

Finally, to investigate organizations' ambidextrous behaviors from a different perspective, we plotted the predicted probabilities of becoming ambidextrous, calculated using the estimated transition probabilities from the other states to the balancing state. Fig. 7 shows that the

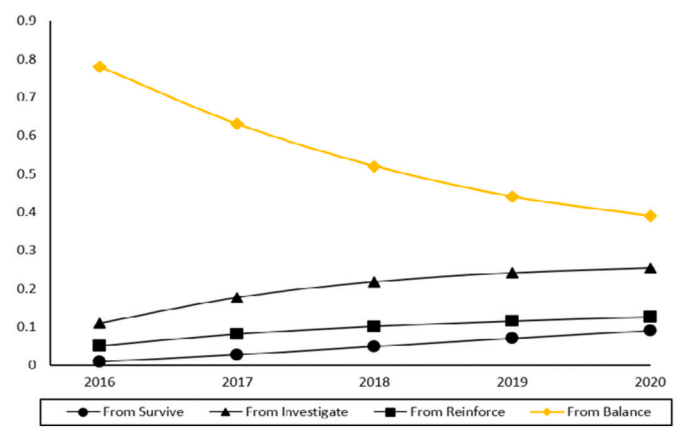

Fig. 7. Predicted probability of being ambidextrous.

probability of staying ambidextrous decreases over time. Further, the probability of reaching ambidexterity from the investigating activities is much higher than the other two types of activities. Collectively, this suggests that firms do not always remain ambidextrous. They need to make a temporary transitions to engage in exclusive periods of exploration and exploitation to become ambidextrous again.

4.2. *Post hoc analysis of the prior security breach effects*

To gain deeper insights, a post hoc analysis was conducted to understand the effects of prior cyberattacks on organizational exploration and exploitation behaviors. The Privacy Rights Clearinghouse's database of data breaches[3] was accessed to gather data on cybersecurity breaches previously encountered by the organizations. We searched each of the 87 companies in the database and determined if any of them

[3] https://privacyrights.org/

had security breaches. We found 19 organizations that experienced cybersecurity breaches. Fig. 8 presents the short-term transition probabilities in the four-state model for organizations that dealt with significant data breaches. Fig. 9 shows the short-term transition probabilities for organizations that did not have any prior security breaches. Comparing the transition probabilities from surviving, reinforcing, and balancing states to the investigating state indicates that firms that experienced cybersecurity breaches are more likely to move to the investigating state to in order to investigate and remediate the breach and prevent future breaches.

4.3. Prediction accuracy

In this study, a general state-based prediction accuracy rate was utilized to measure the overall prediction accuracy for each of the states [69]. The state-based general prediction accuracy rate was determined as follows:

$$R_{acc}{}^{(i)} = \frac{1}{n^{(i)}} \sum_{j=1}^{n^{(i)}} \delta\left(s'^{(i)}_j, s^{(i)}_j\right), i = 1, 2, \ldots, k$$

where $s_j^{(i)}$ represents the observed i^{th} state, $s'^{(i)}_j$ is the predicted i^{th} state, $n^{(i)}$ is the total number of observations in the i^{th}state and k is the number of states in the model. The Markov model was fitted to the state sequence for all the organizations in our sample during the first four time periods (ending 2016, 2017, 2018, and 2019). The period ending 2020 was used as out of sample out of time state sequence for model assessment and validation. The results of state-based predictions show that the average prediction accuracy rates are 96.5% and 93.4% across the states in Models I and II respectively indicating that the multistate Markov model could effectively predict the next state of exploration-exploitation.

5. Discussion

5.1. Findings

The objective of this study was to develop a modeling approach that can help describe how organizations transition between exploration and exploitation states in their cybersecurity response portfolio. We formulated a multistate Markov chain model for analysis of organizations' exploration and exploitation behaviors as found in their annual 10-K reports published by organizations. We described the transition processes by which organizations mobilize and coordinate exploration and exploitation efforts. We estimated the transition probabilities at each state for an organization to move to the next possible state by observing the exploration and exploitation behaviors of organizations.

Our study provides several interesting findings of organizational exploration and exploitation behaviors in the cybersecurity domain. First, this study conceptualized organizational cybersecurity exploration and exploitation as continuous processes which involve a series of incremental changes or transitions between varying levels of exploration and exploitation. Such transitions and shifts between periods of exploration and exploitation have a positive impact on firm performance [35].

We identified four possible states (e.g., surviving, investigating, reinforcing, and balancing) to describe the dynamics of exploration and exploitation in organizational cybersecurity response portfolios. We found that organizations are more likely to stay in their current state to maintain a status quo in the short term. However, as time progresses, they tend to change the level of engagement and move to a different state associated with cybersecurity exploration and exploitation.

We viewed balancing between exploration and exploitation as a dynamic process that occurs gradually and incrementally over time. We considered investigating and reinforcing as two important mechanisms to achieve a balance between exploration and exploitation of cybersecurity responses. When organizations are in the investigating (reinforcing) state, they focus exclusively on exploration (exploitation) during that time period. Consequently, their cybersecurity measures, procedures, capabilities, and even organizational structures are geared toward exploration (exploitation).

Prior studies show conflicting results regarding exploration and exploitation. While Luger et al. [70] assumed that firms remain ambidextrous over time and focus on the firm performance under different environmental conditions, our study is in line with a recent study by

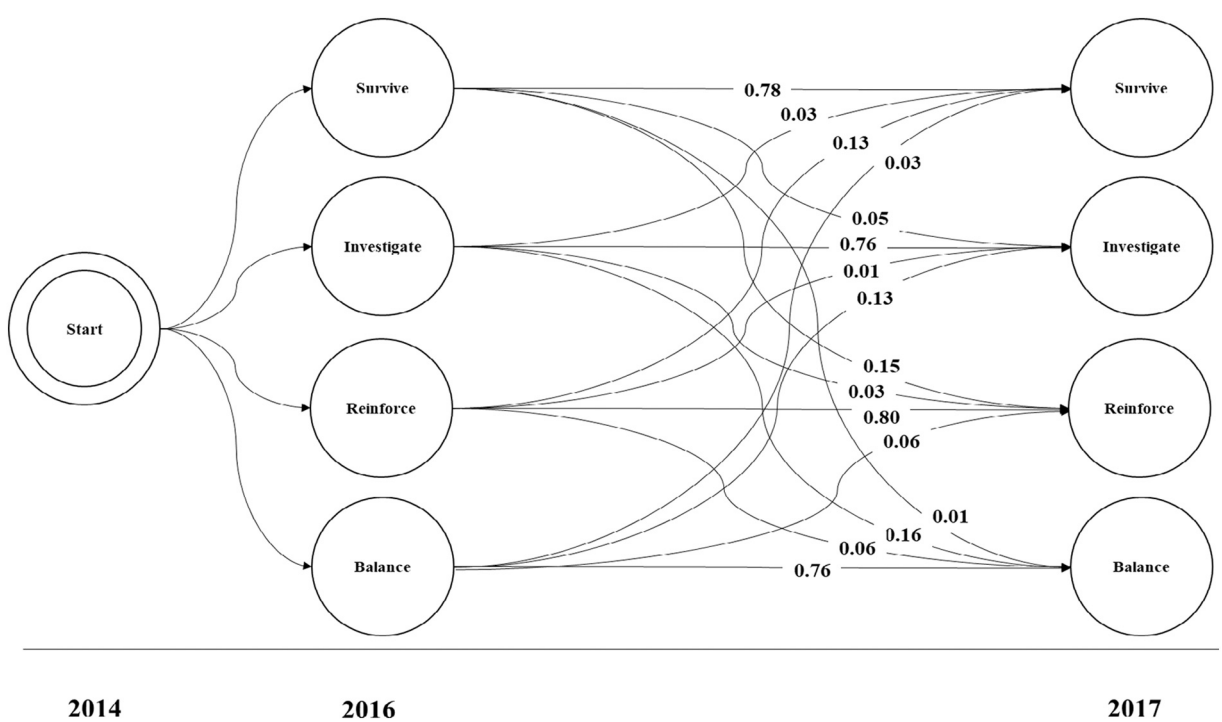

Fig. 8. Short-term transition probabilities for organizations without prior security breaches.

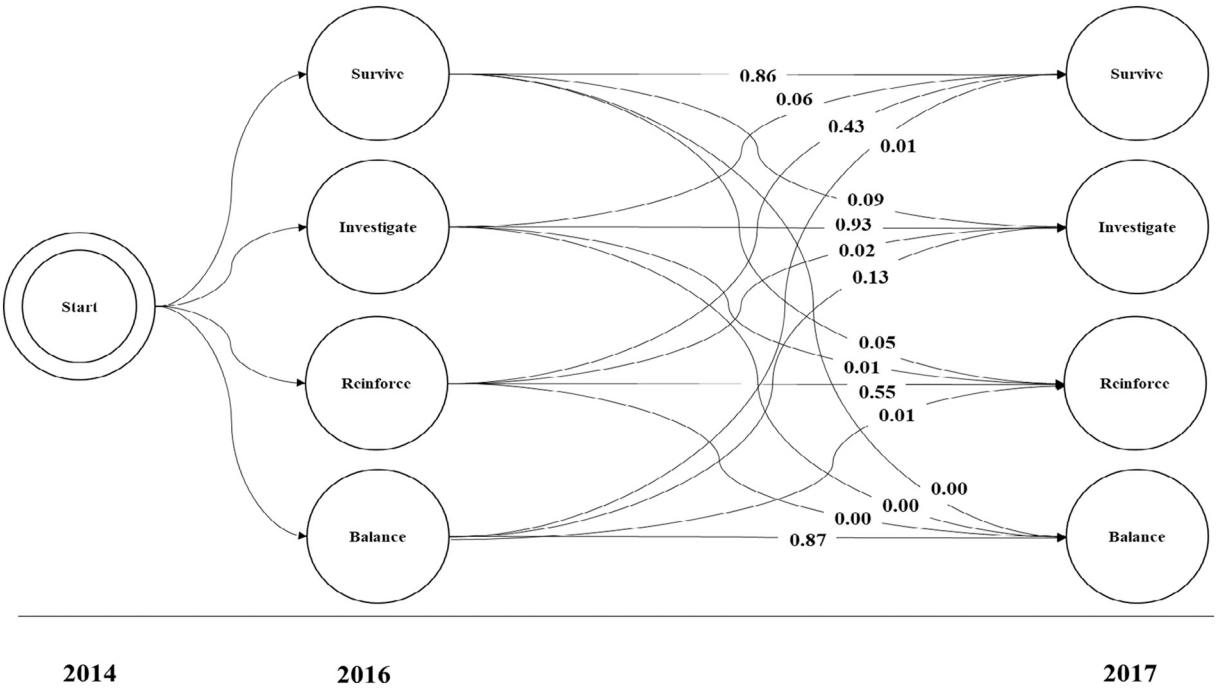

Fig. 9. Short-term transition probabilities for organizations with prior security breaches.

Kang and Kim [35] that organizations do not remain ambidextrous over long periods of time as they need to engage in exclusive periods of exploration and exploitation in order to build their competitive advantage [71].

Our study revealed that organizations spend a fair amount of time in each state for adaptation before leaving that state. This indicates that organizations tend to transition slowly between exploration and exploitation rather than abruptly switch from one to the other. In addition, we positioned surviving and balancing states as two ends of continuum that use exploration and exploitation and found the transition rates between surviving and balancing states are rather low. We argued that organizations do not tend to engage in radical or discontinuous jump from one mode to an opposing mode in their cybersecurity response efforts as it may negatively affect firm performance [35].

Our post hoc analysis on the effects of prior cyberattacks on organizational exploration and exploitation behaviors revealed that organizations that experienced cybersecurity breaches in the past tend to engage in the investigating process to investigate the breach and prevent similar incidents from occurring in the future. This behavior is reflective of a firm's immediate response while handling cyberattacks.

In summary, our findings suggest that organizations need to maintain their long-term focus on both exploration and exploitation to achieve ambidexterity. They should consistently align and realign their exploration and exploitation efforts with the environmental requirements. While it is not possible to predict an optimal path of transitions between exploration and exploitation to attain long-term ambidexterity, organizations need to engage in incremental and gradual transitions between varying levels of exploration and exploitation for survival and competitive advantage.

5.2. Limitations and future research

Our study is not without limitations which open interesting avenues for future research. We recognize that we only observe an organization's level of exploration and exploitation at the time of 10-K filing for each year, and thus our models are only given the observed sequence; however, it is possible that an organization's observed activities are influenced by unobserved activities which were not mentioned in the 10-K annual report. For example, if an organization is in the survival and balancing states at time t *and* $t+1$ respectively, we believe it is necessary to consider that this organization may have been in unobservable states of investigating or reinforcing states before moving to the balancing state. Future research can focus on a continuous-time hidden Markov chain to infer the hidden state path from the observed data [72,73].

This study relied on information in annual 10-K reports to characterize cybersecurity exploration and exploitation activities. However, future studies can consider other primary and benchmark data sources such as interviews, surveys, news, or other corporate disclosures to generalize our findings. Future research could use alternative modeling approaches such as game theory (e.g. Nash dynamic equilibrium) with known or unknown payoff (or reward) structure to understand how organizations handle the trade-offs between exploration and exploitation in their decision process to achieve ambidexterity [74].

In addition, the keywords list compiled in this study to define cybersecurity responses may not be a complete representation of the extent of cybersecurity measures in place at organizations. It may be possible to further refine the list or create an industry-specific list with more detailed classification scheme to represent cybersecurity responses. Therefore, future research can adopt a more diversified approach in gathering terms and phrases used to quantify cybersecurity response. The keyword list can be further improved based on dictionaries containing terms related to hacking, technical, and black-market discussion [75–77].

Finally, future research could apply our Markov models to settings other than cybersecurity. Our sample included S&P-100 large firms, which are typically well-established and viewed as industry leaders. However, it may not be clear to what extent the findings of this study apply to small and medium-sized enterprises (SMEs). Future research could use such operationalizations to study organizational exploration and exploitation in light of the unique characteristics of SMEs.

6. Conclusion

This study proposed a multistate continuous-time Markov chain model to analyze the behavior of organizations as they explore and exploit responses to cybersecurity. This study text-mined data gathered

from the annual reports of organizations to generate longitudinal models to quantify exploration and exploitation efforts by organizations in their cybersecurity responses. The multistate Markov model was used to generate insights into how organizations go through different stages to balance between exploitation and exploration. Given the important role of exploration and exploitation behaviors in organizational learning, this study makes important contributions to the understanding of approaches in which organizations may handle their cybersecurity responses. Our study can serve as the groundwork for further theoretical and analytical development of organizational transition processes between varying levels of exploration and exploitation.

Appendix A

Ticker	Company	Headquarters	GICS Sector	State Sequence
MMM	3 M Company	Minnesota	Industrials	3, 3, 3, 3, 3
ABT	Abbott Laboratories	Illinois	Health Care	3, 3, 3, 3, 3
ABBV	AbbVie Inc.	Illinois	Health Care	3, 3, 3, 3, 3
ACN	Accenture plc	Ireland	Information Technology	4, 4, 4, 4, 4
ALL	Allstate Corp	Illinois	Financials	4, 4, 2, 2, 2
GOOGL	Alphabet Inc	California	Communication Services	4, 4, 4, 4, 4
MO	Altria Group Inc	Virginia	Consumer Staples	2, 2, 2, 2, 2
AMZN	http://Amazon.com Inc.	Washington	Consumer Discretionary	3, 3, 3, 3, 3
AXP	American Express Co	New York	Financials	2, 2, 2, 2, 2
AIG	American International Group, Inc.	New York	Financials	4, 2, 2, 4, 4
AMGN	Amgen Inc.	California	Health Care	3, 1, 2, 2, 2
AAPL	Apple Inc.	California	Information Technology	4, 4, 4, 4, 4
T	AT&T Inc.	Texas	Communication Services	3, 1, 1, 1, 1
BAC	Bank of America Corp	North Carolina	Financials	2, 2, 2, 2, 2
BRK·B	Berkshire Hathaway	Nebraska	Financials	1, 1, 3, 3, 3
BIIB	Biogen Inc.	Massachusetts	Health Care	4, 3, 4, 1, 2
BA	Boeing Company	Illinois	Industrials	3, 3, 3, 3, 3
BMY	Bristol-Myers Squibb	New York	Health Care	1, 2, 2, 3, 4
COF	Capital One Financial	Virginia	Financials	2, 4, 4, 4, 4
CAT	Caterpillar Inc.	Illinois	Industrials	1, 1, 1, 1, 1
CVX	Chevron Corp.	California	Energy	1, 3, 3, 3, 3
CSCO	Cisco Systems	California	Information Technology	3, 3, 3, 3, 3
C	Citigroup Inc.	New York	Financials	3, 3, 1, 1, 1
KO	Coca-Cola Company (The)	Georgia	Consumer Staples	3, 3, 3, 3, 3
CL	Colgate-Palmolive	New York	Consumer Staples	4, 4, 4, 4, 4
CMCSA	Comcast Corp.	Pennsylvania	Communication Services	2, 1, 2, 2, 2
COP	ConocoPhillips	Texas	Energy	3, 3, 3, 3, 1
COST	Costco Wholesale Corp.	Washington	Consumer Staples	3, 3, 3, 3, 1
CVS	CVS Health	Rhode Island	Health Care	2, 2, 2, 2, 2
DWDP	DowDuPont	Michigan	Materials	1, 1, 1, 1, 1
EMR	Emerson Electric Company	Missouri	Industrials	3, 4, 2, 2, 4
EXC	Exelon Corp.	Illinois	Utilities	2, 2, 2, 4, 4
XOM	Exxon Mobil Corp.	Texas	Energy	3, 3, 3, 3, 3
FB	Facebook, Inc.	California	Communication Services	2, 2, 2, 2, 2
FDX	FedEx Corporation	Tennessee	Industrials	3, 1, 1, 1, 1
F	Ford Motor	Michigan	Consumer Discretionary	4, 4, 4, 4, 4
GD	General Dynamics	Virginia	Industrials	4, 4, 4, 4, 2
GE	General Electric[1]	Massachusetts	Industrials	1, 1, 1, 1, 1
GM	General Motors	Michigan	Consumer Discretionary	2, 2, 1, 1, 1
GILD	Gilead Sciences	California	Health Care	2, 4, 2, 2, 2
GS	Goldman Sachs Group	New York	Financials	2, 2, 2, 2, 4
HAL	Halliburton Co.	Texas	Energy	3, 3, 3, 3, 3
HD	Home Depot	Georgia	Consumer Discretionary	3, 3, 3, 3, 3
HON	Honeywell Int'l Inc.	New Jersey	Industrials	4, 4, 2, 2, 2
INTC	Intel Corp.	California	Information Technology	4, 3, 3, 1, 1
IBM	International Business Machines	New York	Information Technology	2, 2, 2, 2, 2
JNJ	Johnson & Johnson	New Jersey	Health Care	3, 1, 1, 1, 1
JPM	JPMorgan Chase & Co.	New York	Financials	2, 2, 2, 2, 1
LLY	Lilly (Eli) & Co.	Indiana	Health Care	4, 1, 3, 4, 3
LMT	Lockheed Martin Corp.	Maryland	Industrials	2, 2, 3, 1, 1
LOW	Lowe's Cos.	North Carolina	Consumer Discretionary	2, 4, 4, 4, 4
MA	Mastercard Inc.	New York	Information Technology	2, 2, 2, 1, 2
MCD	McDonald's Corp.	Illinois	Consumer Discretionary	1, 1, 3, 1, 3
MDT	Medtronic plc[2]	Ireland	Health Care	2, 2, 2, 2, 2
MRK	Merck & Co.	New Jersey	Health Care	4, 4, 4, 4, 4
MET	MetLife Inc.	New York	Financials	4, 4, 4, 4, 4
MSFT	Microsoft Corp.	Washington	Information Technology	4, 4, 2, 2, 2
MDLZ	Mondelez International	Illinois	Consumer Staples	3, 1, 1, 1, 1
MS	Morgan Stanley	New York	Financials	4, 2, 2, 2, 2
NKE	Nike	Oregon	Consumer Discretionary	3, 3, 3, 3, 3
OXY	Occidental Petroleum	California	Energy	1, 1, 1, 1, 1
ORCL	Oracle Corp.	California	Information Technology	4, 4, 4, 4, 4
PEP	PepsiCo Inc.	New York	Consumer Staples	2, 2, 2, 2, 2
PFE	Pfizer Inc.	New York	Health Care	2, 2, 4, 4, 4

(*continued on next page*)

(*continued*)

Ticker	Company	Headquarters	GICS Sector	State Sequence
PM	Philip Morris International	New York	Consumer Staples	3, 3, 1, 3, 1
PG	Procter & Gamble	Ohio	Consumer Staples	1, 1, 1, 1, 1
QCOM	QUALCOMM Inc.	California	Information Technology	3, 1, 3, 3, 3
RTN	Raytheon Co.	Massachusetts	Industrials	2, 2, 2, 4, 4
SLB	Schlumberger Ltd.	Netherlands	Energy	1, 3, 3, 3, 3
SPG	Simon Property Group Inc	Indiana	Real Estate	1, 1, 3, 1, 1
SO	Southern Co.	Georgia	Utilities	4, 3, 4, 4, 4
SBUX	Starbucks Corp.	Washington	Consumer Discretionary	3, 3, 1, 3, 1
TGT	Target Corp.	Minnesota	Consumer Discretionary	3, 1, 1, 1, 1
TXN	Texas Instruments	Texas	Information Technology	3, 3, 3, 3, 3
BK	The Bank of New York Mellon Corp.	New York	Financials	1, 1, 1, 1, 1
DIS	The Walt Disney Company	California	Communication Services	3, 3, 3, 3, 3
FOXA	Twenty-First Century Fox	New York	Communication Services	2, 2, 4, 2, 2
USB	U.S. Bancorp	Minnesota	Financials	3, 3, 3, 3, 3
UNP	Union Pacific	Nebraska	Industrials	1, 1, 1, 1, 1
UNH	United Health Group Inc.	Minnesota	Health Care	3, 4, 4, 4, 4
UPS	United Parcel Service	Georgia	Industrials	4, 4, 2, 2, 2
UTX	United Technologies	Connecticut	Industrials	3, 3, 3, 3, 3
VZ	Verizon Communications	New York	Communication Services	1, 1, 1, 1, 1
V	Visa Inc.	California	Information Technology	1, 1, 2, 2, 2
WBA	Walgreens Boots Alliance	Illinois	Consumer Staples	4, 4, 4, 4, 4
WMT	Walmart	Arkansas	Consumer Staples	4, 4, 2, 2, 2
WFC	Wells Fargo	California	Financials	1, 1, 1, 1, 1

Each cell in the State column shows the state occupied by the organization in each of the time periods: 2014–2016, 2014–2017, 2014–2018, 2014–2019, and 2014–2020 respectively.

Appendix B

Model I – Exploration-Exploitation Transition Probabilities by Industry Sector

GICS Industry Sector		One Year		Five Year		Inter-Transition Probability*	
		Transition Probabilities		Transition Probabilities			
	State	Explore	Exploit	Explore	Exploit	One-Year	Five-Year
Financials	Explore	0.833335	0.166665	0.42799	0.57201	0.18939	0.65000
	Exploit	0.022726	0.977274	0.077999	0.922001		
Information Technology	Explore	0.952381	0.047619	0.783526	0.216474	0.04762	0.21648
	Exploit	1.27E-06	0.999999	5.79E-06	0.999994		
Health Care	Explore	0.739132	0.260868	0.397241	0.602759	0.39879	0.92145
	Exploit	0.137929	0.862071	0.318698	0.681302		
Consumer Discretionary	Explore	0.92	0.08	0.761418	0.238582	0.26181	0.78081
	Exploit	0.181817	0.818183	0.542229	0.457771		
Industrials	Explore	0.800001	0.199999	0.417374	0.582626	0.27407	0.79841
	Exploit	0.074073	0.925927	0.215786	0.784214		
Consumer Staples	Explore	0.739131	0.260869	0.437147	0.562853	0.43734	0.94360
	Exploit	0.17647	0.82353	0.380753	0.619247		
Communication Services	Explore	0.8	0.2	0.510126	0.489874	0.36666	0.89810
	Exploit	0.166667	0.833333	0.408229	0.591771		
Energy	Explore	0.999994	5.99E-06	0.99998	2.01E-05	0.20001	0.67234
	Exploit	0.200006	0.799994	0.672325	0.327675		
Utilities	Explore	0.5	0.5	0.253086	0.746914	0.666667	0.99588
	Exploit	0.166667	0.833333	0.248971	0.751029		

* The Inter-Transition Probability is the transition ratio between exploration and exploitation.

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